

2	(a) scalar has magnitude only vector has magnitude and direction	B1	B1	[2]
(b)	(i) $v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2}mv^2 = mgh$) $v = 22(.1) \text{ m s}^{-1}$	C1	A1	[2]
	(ii) $22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$) $t (=22.1/9.81) = 2.26 \text{ s}$ or $t [(5.097)^{1/2}] = 2.26 \text{ s}$	M1	A0	[1]
	(iii) horizontal distance = $15 \times t$ $= 15 \times 2.257 = 33.86$ (allow $15 \times 2.3 = 34.5$)	C1		
	(displacement) ² = (horizontal distance) ² + (vertical distance) ² $= (25)^2 + (33.86)^2$	C1	C1	
	displacement = 42 (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.)	A1		[4]
	(iv) distance is the actual (curved) path followed by ball displacement is the straight line/minimum distance P to Q	B1	B1	[2]